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Assessing reliable measurements of total organic fluorine in food packaging materials

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ABSTRACT

Per- and polyfluoroalkyl substances (PFAS) are a family of more than 15.000 substances, some of which are considered persistent environmental contaminants. Certain PFAS were commonly used in food contact materials (FCMs) to increase technical performance. Some non-polymeric PFAS species have been found to migrate into food and contribute to human and environmental exposure. Regulatory frameworks, such as the EU Packaging and Packaging Waste Regulation (PPWR) and Danish Executive Order No. 681 from 2020, are increasingly focusing on total fluorine (TF) and total organic fluorine (TOF) thresholds as proxies to control PFAS presence in packaging. However, current analytical methods face limitations in accurately quantifying TOF, particularly in complex matrices containing both organic and inorganic fluorine, generating false results. This study presents a robust method for indirect TOF determination in solid matrices, combining combustion ion chromatography (CIC) with an optimised pre-combustion step at 525 °C for 2 h. The method was validated using a diverse set of fluorinated compounds, including polymeric and non-polymeric organic fluorine, and tested on mineral fillers used in paper manufacturing. Compared to other pre-treatment methods for indirect TOF determination, aqueous extraction used to remove inorganic fluorine often overestimates TOF, likely because of its limited ability to fully extract inorganic fluorides. The proposed method minimises false positives and enhances the reliability of TOF quantification, supporting regulatory compliance and environmental monitoring. These findings underscore the importance of accurate TOF assessment in FCMs and highlight the need for harmonised analytical approaches to better evaluate organic fluorine contamination in consumer products, and their environmental implications.

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Total organic fluorine; combustion ion chromatography; food contact materials; pre-combustion; PFAS

Introduction

Per- and polyfluoroalkyl substances (PFAS) constitute a vast class of over 15,000 synthetic fluorinated compounds, categorised as either polymeric or non-polymeric (Chang et al. 2025, Liu et al. 2025). These substances are extensively employed in consumer and industrial applications due to their hydrophobic and lipophobic properties, which confer resistance to water, grease, and stains. Consequently, PFAS are commonly found in textiles, paper products, and food contact materials (FCMs) (Pervez et al. 2025, Ramírez Carnero et al. 2021).

The environmental persistence and bioaccumulative nature of PFAS have led to their designation as "forever chemicals" (Sunderland et al. 2019). While legacy PFAS have been extensively characterised, emerging compounds remain poorly documented, posing challenges for risk assessment and regulatory oversight (Brase et al. 2021).

Human exposure to PFAS occurs through multiple pathways, with dietary intake and drinking water identified as primary sources (De Silva et al. 2021). PFAS contamination has been reported in various food items, such as fish, meat, seafood, and packaged products like microwave popcorn (Torres and De-la-Torre 2023; Guzman-Puyol 2024). A significant contributor to this contamination is the migration of PFAS from FCMs into food matrices (Ramírez Carnero et al. 2021; Phelps et al. 2024).

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In response to growing human and environmental exposure, regulatory frameworks have been implemented globally. In the United States, several states have enacted restrictions on PFAS in food packaging (Saldaña-Pierard et al. 2025). Within the European Union, the proposed Packaging and Packaging Waste Regulation (PPWR) prohibits the intentional use of PFAS, establishing maximum concentration limits of 25 µg/kg for individual non-polymeric PFAS and 250 µg/kg for their total content (European Commission 2025). Additionally, a threshold of 50 mg/kg for total fluorine (TF) has been proposed, encompassing both polymeric and non-polymeric PFAS (European Commission 2025). Denmark has further restricted the use of fluorinated substances in paper and board FCMs lacking functional barriers, with an indicative limit of 20 µg organic fluorine per gram of paper (Danish Veterinary and Food Administration 2020).

Given the complexity and diversity of PFAS structures, comprehensive characterisation in FCMs remains analytically challenging. Conventional targeted methods, such as liquid chromatography-tandem mass spectrometry (LC-MS/MS) for ionic PFAS and gas chromatography-mass spectrometry (GC-MS) for semi-volatile PFAS, are limited in scope (Androulakakis et al. 2022; Schiavone and Portesi 2023). Even advanced techniques like high-resolution mass spectrometry (HRMS) used for suspect and non-target screening are constrained by the lack of authentic standards, impeding definitive identification and quantification.

To address these limitations, alternative approaches have been developed to assess total fluorine (TF) content in FCMs. Techniques such as particle-induced gamma-ray emission (PIGE) (Schaider et al. 2017), instrumental neutron activation analysis (INAA) (Schultes et al. 2019), fluorine ion-selective electrode (F-ISE) coupled with oxygen bomb combustion (Ignacio et al. 2023), and combustion ion chromatography (CIC) have been explored (Jovanović et al. 2024). Among these, CIC has emerged as the most widely adopted method due to its superior sensitivity and its capability to analyse the entire sample, thereby providing a whole-sample TF measurement (Schultes et al. 2019). In contrast, PIGE primarily probes the surface layer (penetration depth ≈ 0.22 mm), which can lead to overestimation in coated or non-homogeneous materials. For instance, Schultes et al. (2019) reported higher TF values by PIGE for thick paperboard, attributed to surface coating heterogeneity. INAA, while non-destructive, may be limited by matrix composition. In the same study, BCR-461 (a certified reference material for fluorine in clay powder) could not be analysed by INAA because of its high aluminium content. Detection limits further differentiate these methods: CIC typically achieves the lowest LOD (~1 mg/kg), compared with INAA (~20 mg/kg) and PIGE (~38 mg/kg), making CIC the most sensitive approach. These considerations underscore the complementarity of these techniques and support the selection of CIC for TF determination, while PIGE and INAA remain useful for rapid screening depending on sample characteristics. However, CIC does not inherently distinguish between organic and inorganic fluorine, necessitating appropriate sample preparation to enable TOF determination (Idjaton et al. 2024). The need for sample preparation is particularly critical due to the lack of harmonised methodologies for TOF determination in solid materials, which leads to considerable variability in results, especially in matrices containing both organic and inorganic fluorine (Omer et al. 2025).

To differentiate between organic and inorganic fluorine, several pre-treatment strategies have been proposed. One approach involves extracting the organic fluorine fraction using organic solvents (e.g. MeOH), allowing CIC to quantify extractable organic fluorine (EOF) (Idjaton et al. 2024; Jovanović et al. 2024). However, this approach is not applicable to fluorinated compounds that are insoluble in organic solvents such as MeOH, including polymeric PFAS characterised by high molecular weight (Roesch et al. 2024). An alternative strategy involves dual analysis: one sample is used to determine TF, while the other is treated to quantify total inorganic fluorine (TIF). TOF is then calculated as the difference between TF and TIF (Hu et al. 2025; Ignacio et al. 2023). TIF can be estimated by sonicating the sample in water and analysing the supernatant using F-ISE or ion chromatography (IC) (Hu et al. 2025; Ignacio et al. 2023). This method assumes that inorganic fluorides are water-soluble and organic fluorides are not.

However, this assumption may lead to inaccuracies: poorly soluble inorganic salts (e.g. CaF_2, MgF_2, MnF_2, AlF_3) or inorganic fluorine trapped in fillers (talc, kaolin) may result in underestimation of TIF (Savenko and Savenko 2023; Curtzwiler et al. 2025). Another method for TIF determination consists in thermally treating the sample to convert it to ash (removal of the organic fraction), as described in ISO 1762 (ISO 2019) and EN 13432 (European Committee for Standardization 2000). Although these standards do not refer directly to fluorinated compounds, they define inorganic matter as the residue after calcination at 525–550 °C of paper, cardboard and packaging recoverable through composting and biodegradation (European Committee for Standardization 2000; ISO 2019).

This study aims to develop and validate a robust analytical strategy for the indirect quantification of TOF in solid matrices, particularly FCMs. The proposed method integrates CIC with a pre-combustion step designed to eliminate organic fluorine while preserving inorganic residues. The approach is evaluated using a range of fluorinated compounds with diverse physicochemical properties, and its applicability is assessed through comparative analysis with the aqueous extraction technique.

Materials and methods

Chemicals, reagents and standard solutions

All chemicals, reagents, and standard solutions were of analytical grade. Specific details regarding the compounds used for calibration and validation are provided in the supplementary material.

Instrumentation and IC conditions

Two combustion systems were employed for the determination of total fluorine (TF) *via* combustion ion chromatography (CIC):

- System 1: AQF-2100 (Mitsubishi Environsciences, Düsseldorf, Germany), comprised an ASC-2700LS autosampler, an ABC-210 automatic boat controller, an HF-210 horizontal oven, and a GA-211 sample absorption unit. The initial combustion method (Initial method 1) provided by the manufacturer is described in Supplemental Table S1.
- System 2: Xprep CIC (Trace Elemental Instruments, Houston, USA), included a combustion module equipped with a Newton autosampler and a fraction collection module. The initial combustion method (Initial method 2) supplied by the manufacturer is detailed in Supplemental Table S1.

The initial combustion methods were optimised to improve TF recovery. The final combustion method (Final optimised method) is detailed in Supplemental Table S2.

Ion chromatography (IC) analyses were conducted using a system equipped with a KOH eluent generator cartridge (Dionex EGC 500) and a trap column (Dionex CR-ATC 600) to maintain eluent purity. A temperature-controlled column oven ensured consistent separation conditions. Separation was achieved using a Dionex AS28 analytical column (4 µm, 2 × 150 mm) preceded by a Dionex AG28 guard column (4 µm, 2 × 30 mm). The anion suppressor used was an Dionex ADRS 600 dynamically regenerated electrolytic suppressor (2 mm diameter) operated in constant voltage mode. Detection was ensured with a conductivity detector. The IC method parameters are summarised in Table S3.

Approach for fluorine fraction analysis

Total fluorine (TF)

The TF concentration was measured directly on samples without any pre-treatment. The total fluorine yield (%) was calculated by comparing the measured TF concentration with the theoretical fluorine content derived from the molecular formula and purity of each compound.

$$\text{TF Yield}(\%) = \left(\frac{\text{Measured TF}(\text{mg/kg})}{\text{Theoretical TF}(\text{mg/kg})} \right) \times 100$$

Total inorganic fluorine (TIF)

The TIF concentration was determined by subjecting the sample to a pre-combustion step (using muffle furnace) designed to volatilise

organic fluorine. The remaining inorganic residue (ash) was then analysed by CIC. TIF was expressed as a percentage of TF.

$$\mathrm{TIF}(\%\mathrm{of\,TF}) = \begin{pmatrix} \mathrm{Measured\,TIF}(\mathrm{mg/kg})/ \\ \mathrm{Measured\,TF}(\mathrm{mg/kg}) \end{pmatrix} \times 100$$

Total organic fluorine (TOF)

TOF was calculated indirectly by subtracting the TIF value from the TF value. This approach assumes complete volatilisation of organic fluorine and retention of inorganic fluorine during pre-combustion.

$$\mathrm{TOF}(\mathrm{mg/kg}) = \mathrm{Measured\,TF}(\mathrm{mg/kg}) - \mathrm{Measured\,TIF}(\mathrm{mg/kg})$$

$$\mathrm{TOF}(\%\ \mathrm{of\ TF}) = \begin{pmatrix} \mathrm{TOF}(\mathrm{mg/kg})/ \\ \mathrm{Measured\ TF}(\mathrm{mg/kg}) \end{pmatrix} \times 100$$

Selection of the model compounds

To evaluate TF recovery and optimise the pre-combustion step for indirect TOF determination, a broad range of fluorine-containing substances (inorganic salts and organic materials) were selected based on their contrasting physicochemical properties (Supplemental Table S4). All of these salts are analytical standards and were purchased from ThermoFisher Scientific (Darmstadt, Germany).

Six inorganic salts [ammonium tetrafluoroborate (NH_4BF_4), potassium tetrafluoroborate (KBF_4), sodium hexafluorosilicate (Na_2SiF_6), calcium fluoride (CaF_2), zinc fluoride (ZnF_2) and sodium fluoride (NaF)] were chosen to represent compounds with diverse thermal stabilities and melting points. This selection ensures that the method is tested against refractory fluorides as well as more volatile species.

Polytetrafluoroethylene (PTFE) was chosen as the reference organic fluorinated compound due to its widespread application and high thermal stability (Dhanumalayan and Joshi 2018). PTFE was analysed in both powder and ribbon forms to assess the consistency of the pre-combustion process across different physical states. The underlying hypothesis was that complete volatilisation of PTFE under selected conditions would ensure the removal of less thermally stable organic fluorinated compounds. Two low-molecular-weight organic fluorinated compounds, 2,5-difluorobenzenesulfonamide and 4-fluorobenzenesulfonamide, were also incorporated to represent less thermally stable organic species.

In addition, two certified reference materials (CRMs) from the Joint Research Centre (European Commission), BCR-460 (fluorine in coal powder (225 ± 8 mgF/kg)) and BCR-461 (fluorine in clay powder (568 ± 60 mgF/kg)), were included to provide well-characterised benchmarks for method validation. These CRMs are commonly used in environmental and materials analysis and offer a controlled matrix for assessing the accuracy and reliability of TF determination.

Methodological pre-treatment comparison for indirect TOF determination: pre-combustion vs. aqueous extraction

To evaluate the discrimination and robustness of the pre-combustion method for indirect TOF determination, a series of mineral fillers were selected. The mineral fillers analysed (talc, kaolin, silica, and calcium carbonate) are widely used in paper manufacturing to enhance opacity and printability (Hubbe and Gill 2016). Each sample was analysed using two approaches for indirect TOF determination.

TIF determination using pre-combustion

For each sample, six combustion boats (representing three replicates for each condition to assess repeatability) were prepared with approximately 50 mg of sample.

- Three boats were directly analysed by CIC to determine the TF content.
- Three boats were subjected to heat treatment in a muffle furnace to determine the total TIF under optimised pre-combustion conditions (525 °C for 2 hours).

- The TOF was then calculated by subtracting the average TIF value of each approach from the average TF value.

TIF determination using aqueous extraction

To compare the pre-combustion-based approach with an extraction method, the same materials were also subjected to aqueous extraction for TIF determination. The protocol was as follows:

- 1 g of finely ground sample was weighed into a 50 mL centrifuge tube and mixed with 10 mL of Milli-Q water.
- The mixture was sonicated in a water bath for 10 minutes, then centrifuged at 4500 rpm for 15 minutes.
- The supernatant was collected in a 15 mL polypropylene vial and directly injected into the ion chromatograph to quantify the water-soluble inorganic fluorine fraction (TIF).

Quality assurance and quality control (QA/QC)

A detailed description of the QA/QC is provided in the (Supplementary material Table S5).

Results and discussion

Optimisation of total fluorine (TF) determination by combustion ion chromatography (CIC)

The initial combustion methods provided by the suppliers on both systems were evaluated. These methods differed in two key parameters: the combustion temperature and the residence time of the sample boat in the high-temperature zone (also called the END zone). Initial method 1 consisted of combustion at 1000 °C with a 90-second residence time in the END zone, whereas Initial method 2 consisted of combustion at 1050 °C with a 360-second residence time in the END zone. These differences can be explained by the different analytical objectives of each method. Initial method 1 was developed for polymer matrices, where fluorine is predominantly present in organic form and decomposes more readily. In contrast, Initial method 2 was designed for a broader range of materials, including paper and multilayer packaging, which may contain thermally stable inorganic fluorides.

To assess the performance of each method, a selection of fluorinated compounds (Supplemental Table S3) was analysed. As shown in Figure 1, a complete fluorine recovery was observed for

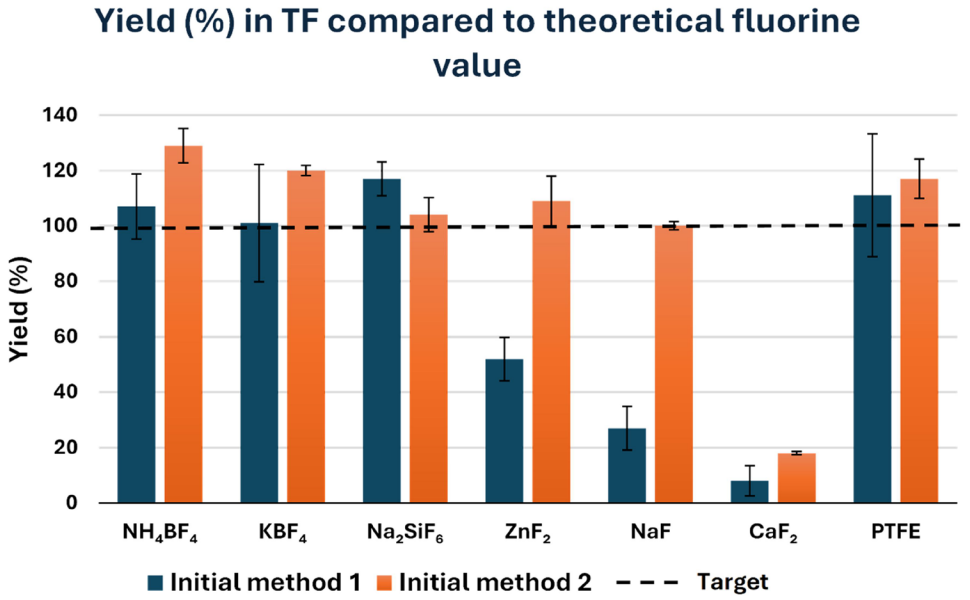


Figure 1. Comparison of TF yields (%) with theoretical fluorine values for various compounds using the methods provided of both systems. Error bars represent standard deviation (Mean ± SD, $n = 3$). The theoretical fluorine value was calculated based on each compound's molecular formula and product purity.

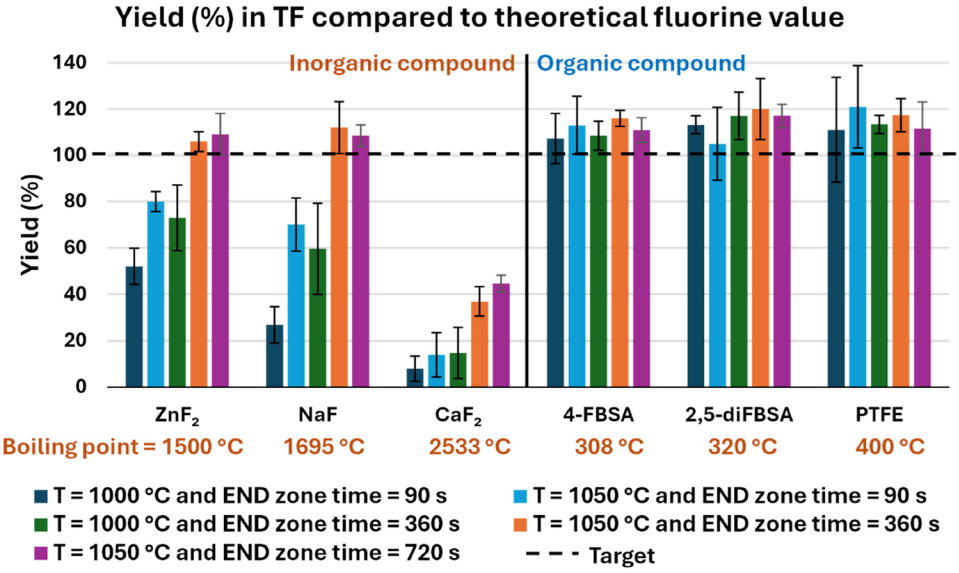


Figure 2. TF yield (%) obtained by increasing combustion temperature and/or residence time in the END zone. Error bars represent standard deviation (Mean ± SD, $n = 3$). The theoretical fluorine content for each compound was calculated from its molecular formula and product purity.

compounds with boiling points below 1000 °C regardless of the method used. However, significant discrepancies were noted for compounds with higher boiling points, particularly for Initial method 1. These deviations could be attributed to insufficient combustion temperature and residence time in the END zone, resulting in incomplete volatilisation of refractory inorganic fluorides.

To improve TF recovery obtained with Initial method 1, the combustion temperature increased to 1050 °C and the residence time was extended to 360 s in the END zone. The impact of each parameter was first individually tested and then in combination using CaF_2, NaF, ZnF_2 and PTFE.

For the inorganic salts, the increase in combustion temperature from 1000 °C to 1050 °C, while maintaining a fixed boat residence time of 90 s, resulted in higher TF yields, particularly for ZnF_2 and NaF, as illustrated in Figure 2. Although CaF_2 also showed an increase in TF yield, its recovery remained low due to its high thermal stability. Extending the boat residence time in the END zone from 90 to 360 s at 1000 °C similarly enhanced TF yields for ZnF_2 and NaF. Despite these improvements, neither parameter alone was sufficient to achieve complete combustion. However, when both the combustion temperature and residence time were simultaneously increased, TF yields for ZnF_2 and NaF reached complete combustion. In contrast, CaF_2 continued to exhibit incomplete combustion even with a residence time of 720 s, underscoring its resistance to thermal degradation. This issue could be partially mitigated by the addition of combustion aids such as tungsten trioxide (WO_3), as recommended in DIN EN 62321-3-2:2016-08 to improve halogen recovery from refractory materials. However, this approach was excluded in this study to preserve the simplicity and reproducibility of the method. Due to the extended analysis time and limited gain of residence of 720 s, the 1050 °C/360 s protocol was adopted for both systems.

Unlike inorganic salts, PTFE exhibited complete combustion regardless of the temperature and residence time in the END zone (Figure 2). Other organic fluorinated compounds (4-fluorobenzenesulfonamide (4-FBSA) and 2,5-difluorobenzenesulfonamide (4-diFBSA) analysed in this study also demonstrated complete combustion across all conditions, confirming that incomplete combustion issues are primarily associated with inorganic fluorinated compounds, particularly those with high thermal stability and low volatility. In contrast, organic fluorinated molecules tend to decompose more easily under the effect of heat, which facilitates the release of fluorine during combustion. Importantly, this observation supports the main objective of this study (indirect determination of TOF). If

incomplete volatilisation of inorganic fluorides does not influence TOF determination, the method remains robust and suitable for achieving this goal.

In summary, the Final optimised combustion method (1050 °C with a 360 s residence time) in orange in Figure 2 enabled complete combustion of organic and inorganic fluorides with boiling points below 1700 °C. However, compounds such as CaF_2 remained partially combusted. Consequently, TF values reported using this method should be understood as representative of all materials containing fluorine compounds that are combustible under these specific conditions.

Evaluation of the final optimised method accuracy for TF determination using CRMs

To assess the accuracy of the Final optimised combustion method for TF determination, a comparison was performed between the Initial method 1 and the Final optimised combustion method using certified reference materials (CRMs).

The observations for inorganic salts and PTFE align with the results obtained for certified reference materials (Figure 3). Increasing temperature and residence time enhanced TF recovery for refractory salts, and a similar trend was observed for BCR-461 (clay), which required more severe thermal conditions (Final optimised combustion method) to achieve complete fluorine release. Conversely, BCR-460 (coal) reached full recovery under Initial method 1 (1000 °C, 90 s), reflecting its lower thermal stability. This comparison underscores the strong influence of matrix composition and thermal resistance on combustion efficiency. It also highlights the potential risk of relying solely on a less demanding CRM such as BCR-460, which may lead to the erroneous assumption that the method guarantees complete combustion. Incorporating more challenging materials like BCR-461 is therefore essential for robust validation and for defining optimised parameters that address the limitations observed with highly stable inorganic fluorides.

In conclusion, even though the Final optimised combustion method does not enable complete combustion of highly refractory compounds such as CaF_2, it demonstrates accuracy when tested against certified reference materials, confirming the reliability of the approach for TF determination.

Optimisation of indirect total organic fluorine (TOF) determination

The strategy adopted in this study relies on a pre-combustion step aimed at volatilising the organic fluorine fraction through thermal treatment at a defined temperature and duration.

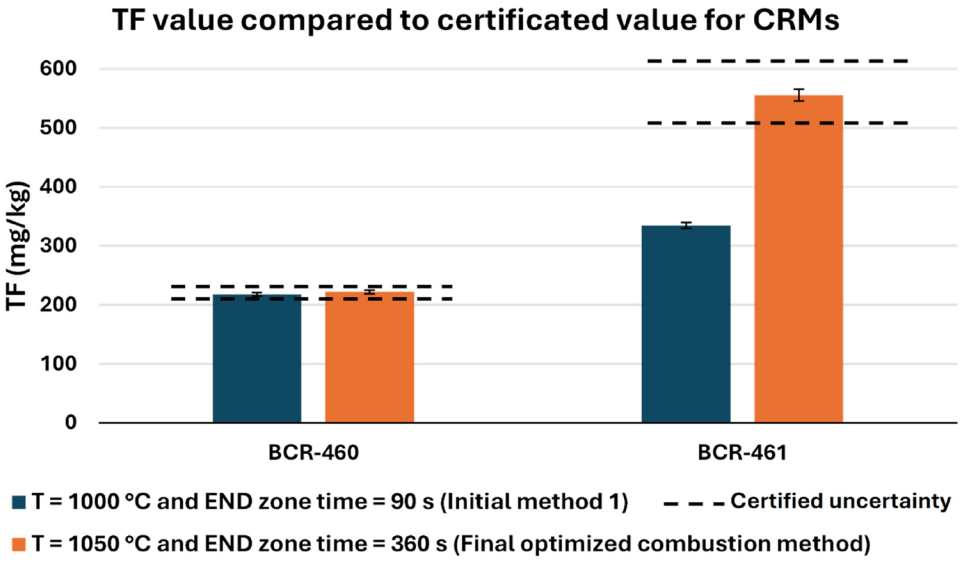


Figure 3. TF (mg/kg) value obtained by comparing *Initial method 1* and *Final optimised combustion method* using CRMs. Error bars represent standard deviation (Mean ± SD, $n = 3$). For the BCR-460 and BCR-461, the certified uncertainty values correspond to the ranges specified in their certificates: 217–233 mg/kg for BCR-460 and 508–628 mg/kg for BCR-461.

Following this thermal treatment, the remaining inorganic residue (ash) is analysed *via* CIC to quantify the TIF. The TOF is then indirectly calculated by subtracting the TIF value from the TF initially measured.

Optimisation of pre-combustion conditions for indirect TOF determination

As a starting point, a reference pre-combustion condition of 550 °C for 2 h was evaluated using PTFE as a model organic fluorinated compound, alongside four inorganic salts. As shown in Figure 4, PTFE exhibited a TOF recovery >99%, confirming its complete volatilisation under these conditions. However, several inorganic salts also volatilised partially or completely, leading to overestimation of TOF values (TOF interferences). As an illustration, KBF_4 subjected to pre-combustion at 550 °C for 2 h yielded only 24% TIF, while the calculated TOF reached 76%, despite the theoretical expectation of 100% TIF. This example highlights that TOF interference generally varies with the thermal stability of the compound: the higher the boiling point and thermal resistance, the lower the percentage of TOF interference. Nevertheless, ZnF_2 displayed unexpectedly high TOF interference (~99%), despite its ionic nature and high boiling point. This anomaly may be attributed to partial covalent behaviour in Zn–F bonds (Yu et al. 1999), emphasising that the relationship between thermal stability and TOF interference is not absolute. These findings highlight the necessity of critically assessing the thermal behaviour of fluorinated compounds within heterogeneous matrices during pre-combustion treatment. This highlights the need for further refinement of the pre-combustion steps to reduce such interferences and improve the selectivity of the method.

To reduce such interferences, a two-step optimisation was performed. The goal was to determine the lowest temperature and shortest time that would ensure complete volatilisation of PTFE, used here as a model compound for thermally stable organic fluorine. This step was critical to avoid false negatives, which could occur if organic fluorine compounds were not fully volatilised during pre-combustion, leading to their misclassification as inorganic fluorine and an underestimation of TOF.

- Step 1: Temperature Optimisation (Fixed Time: 2 h)

Temperatures of 400 °C, 450 °C, and 525 °C were tested. PTFE volatilised completely only at 525 °C (Figure 5(a)).

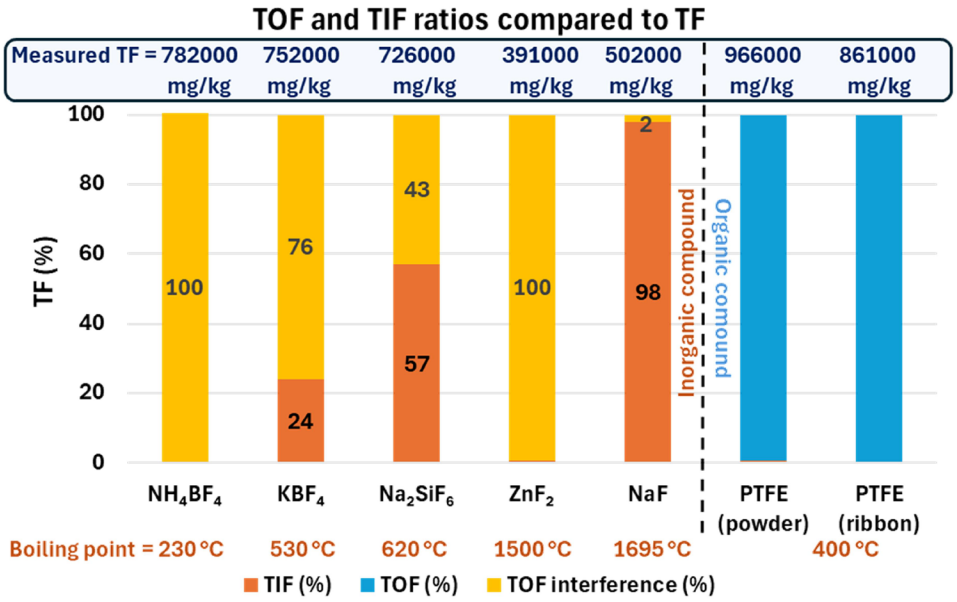


Figure 4. TIF and TOF and TOF interference ratios in selected inorganic salts and PTFE under pre-combustion at 550 °C for 2 h. TOF and TIF and TOF interference (%) are calculated relative to TF measured using the *Final optimised combustion method*. TF and TIF were determined in triplicate ($n = 3$). TOF was determined by subtracting the mean TIF from the mean TF.

- Step 2: Time Optimisation (Fixed Temperature: 525 °C)

Durations of 1 h and 1.5 h yielded TOF recoveries of 84% and 92%, respectively. A duration of 2 h was required for complete volatilisation (Figure 5(b)).

Based on these results, the optimised condition (525 °C for 2 h) was selected for subsequent analyses. Once the optimal pre-combustion condition was found for TOF determination, interferences related to inorganic fluorinated compounds were then evaluated to highlight the limitations of the selected method and to have a critical view of the results obtained.

Verification of the final pre-combustion condition on inorganic compounds

The robustness of the pre-combustion condition was evaluated by comparing TOF recoveries for the same inorganic salts under both reference and final optimised conditions. In this evaluation, only salts that previously exhibited TOF interferences were selected to determine whether the optimised method could effectively reduce these interferences.

As shown in Supplemental Figure S1, similar interference levels were observed, except for Na_2SiF_6, which showed reduced TOF yields under optimised conditions. However, NH_4BF_4 and KBF_4 continued to exhibit high TOF values due to their low melting points, suggesting that interference from these salts is unavoidable if complete PTFE volatilisation is required.

ZnF_2 also maintained extremely high TOF interference (~99%) under both conditions, which contrasts with the expected trend linking thermal stability to reduced interference. This persistent behaviour indicates that factors beyond simple melting point or boiling point influence volatilisation during pre-combustion,

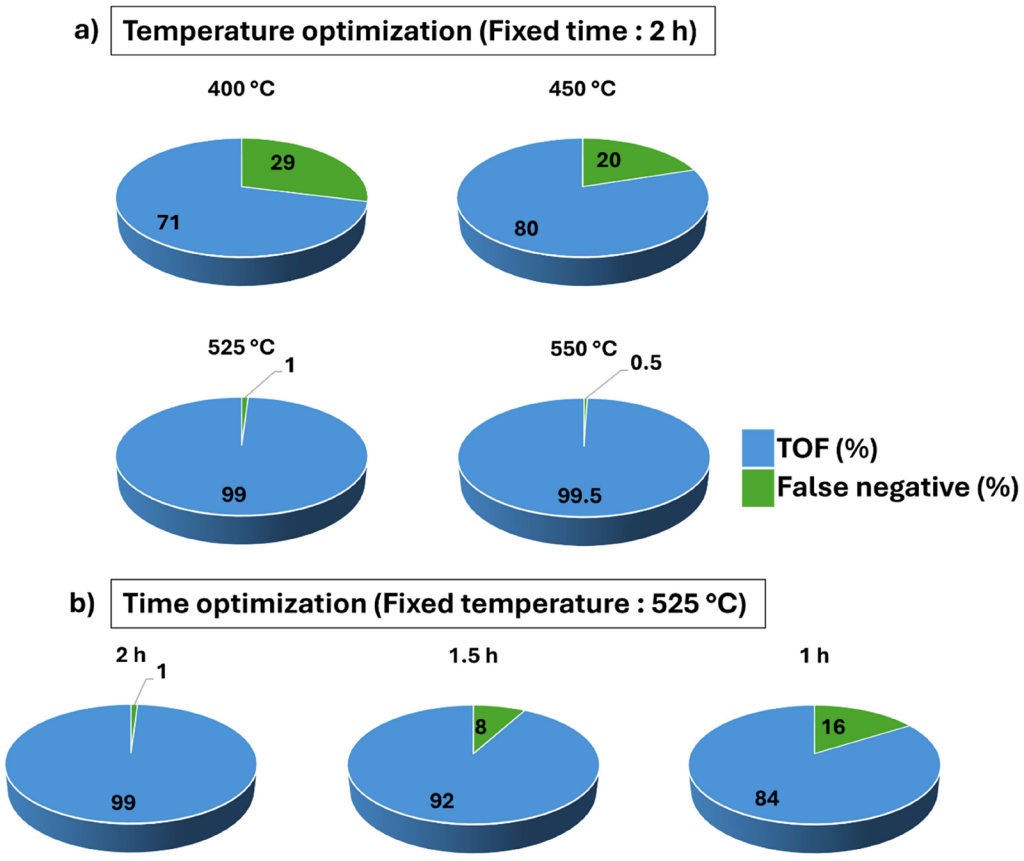


Figure 5. Evaluation of different pre-combustion conditions by varying temperature (a) and time (b) on TOF determination in PTFE powder. TF measured using the *Final optimised combustion method* and TIF were determined in triplicate ($n = 3$). TOF was determined by subtracting the mean TIF from the mean TF. False negatives were identified when TOF values fell below 100% for PTFE, indicating incomplete volatilisation of organic fluorine and underestimation of TOF.

reinforcing the complexity of predicting interference solely based on thermal stability. These findings highlight the necessity of critically assessing the thermal behaviour of fluorinated compounds within heterogeneous matrices during pre-combustion treatment. In real samples, which may contain a mixture of different fluorinated compounds, the assumption that all inorganic fluorides remain stable and non-volatile under pre-combustion conditions may not hold valid. Certain inorganic fluorides can exhibit unexpected partial or complete volatilisation, leading to misclassification as organic fluorine and consequently to overestimation of TOF. Therefore, the indirect determination of TOF must be interpreted with caution, particularly when applied to heterogeneous samples with unknown fluorine composition.

Analysis of mineral fillers used in paper manufacturing

To evaluate the relevance of the optimised pre-combustion condition (525 °C for 2 h), a series of mineral fillers commonly used in paper manufacturing were analysed. The objective was to compare this condition with that suggested by preliminary European normalisation work (700 °C for ≥2 h) to evaluate the impact of more aggressive thermal pre-combustion condition for indirect TOF determination.

The fillers analysed included talc, kaolin, silica, and calcium carbonate. These materials are incorporated into paper to enhance opacity and printability (Hubbe and Gill 2016). Although these fillers are not fluorinated by origin, they are sourced from natural quarries and are not purified analytical standards. Their high incorporation rate (up to 20–30%) of the total paper composition depending on the application makes them a potential source of fluorine contamination due to impurities (Grönfors 2010).

Unexpectedly high TF values were observed in certain fillers, notably talc (973 mg/kg) and kaolin (170 mg/kg), despite their theoretical non-fluorinated nature (Figure 6). These findings suggest the presence of fluorine impurities, the origin of which remains undetermined.

Under the optimised pre-combustion condition (525 °C for 2 h), TF was found to be predominantly composed of TIF, with percentages ranging from 98% to 99.8% for kaolin, calcium carbonate, and talc. This indicates minimal interference in TOF determination under these conditions.

However, a notable difference was observed between powdered and liquid kaolin. The dissolved form exhibited a higher TOF interference, suggesting that dissolved inorganic salts are more susceptible to volatilisation, thereby increasing interference.

When the more aggressive condition (700 °C for 2 h) was applied, interference increased significantly. For example, TOF interference values for powdered kaolin rose from <2% to 86%, indicating substantial overestimation of TOF. While this could suggest non-compliance with certain regulatory thresholds, it is important to clarify that the EU PPWR sets its 50 mg/kg limit on TF, not TOF. Consequently, a TF value above this threshold may be misinterpreted as evidence of PFAS contamination, even when the excess fluorine originates from inorganic sources. This underscores the critical need for well-calibrated methods capable of distinguishing between organic and inorganic fluorine. In addition, Denmark's guideline of 20 mg/kg specifically targets TOF in paper and board FCMs without functional barriers, further emphasising the importance of accurate differentiation to avoid or reduce false positives.

These results underscore the importance of selecting appropriate pre-combustion conditions when analysing materials containing mineral fillers. Overly aggressive thermal treatment may lead to volatilisation of inorganic fluorides, resulting in artificially elevated TOF values and erroneous regulatory conclusions.

Impact of pre-treatment methodologies on indirect TOF determination: pre-combustion vs. aqueous extraction

To assess the reliability and selectivity of the optimised pre-combustion method for indirect determination of TOF, a comparative study was

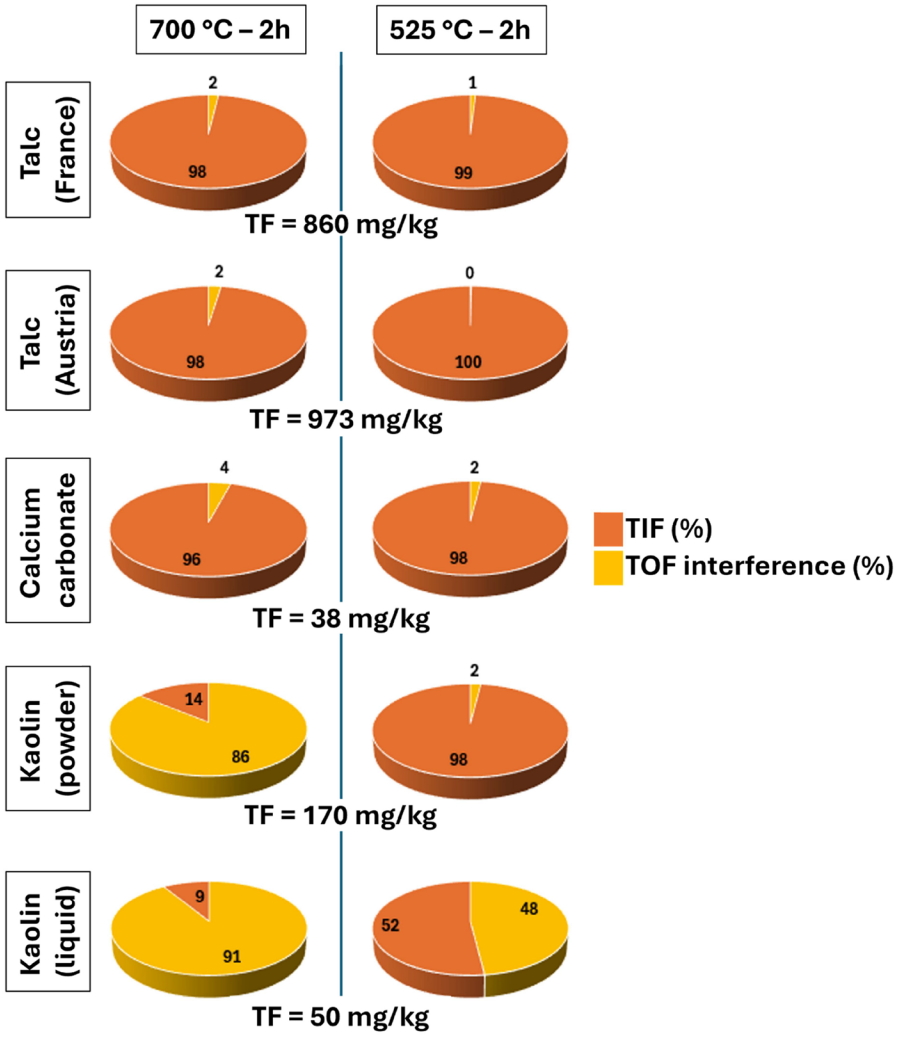


Figure 6. Determination of TF and TIF using two different pre-combustion conditions (700 °C – 2 h and 525 °C – 2 h) in various mineral fillers. TIF and TOF interference fractions (%) are calculated relative to TF measured using the *Final optimised combustion method*. TF measured and TIF were determined in triplicate ($n = 3$). TOF was determined by subtracting the mean TIF from the mean TF.

conducted. Two samples were analysed: talc and powdered kaolin.

Each sample was subjected to two analytical protocols:

1. Pre-combustion at 525 °C for 2 hours, followed by CIC analysis.
2. Aqueous extraction, followed by direct IC analysis of the extract.

The comparative evaluation of pre-combustion and aqueous extraction highlights significant differences in their ability to discriminate between organic and inorganic fluorine (Figure 7). Pre-combustion approach resulted in TIF fractions close to 98–99% of TF, indicating that most of the fluorine present was inorganic. In contrast, aqueous extraction yielded TOF interferences near 99%, suggesting that the method was insufficient to extract inorganic fluorides, which are often poorly soluble or embedded in the matrix. This discrepancy demonstrates that aqueous extraction tends to underestimate TIF and consequently overestimate TOF, leading to potential misclassification of samples as containing high levels of organic fluorine.

Conclusions

A robust and innovative methodology for the indirect determination of TOF in FCMs was developed and validated. By integrating CIC with

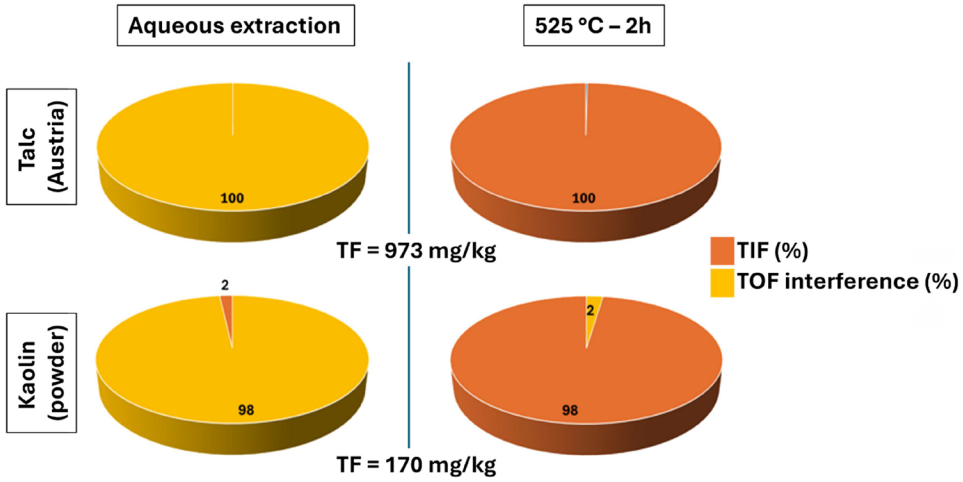


Figure 7. Comparison of pre-combustion and aqueous extraction methods for TIF and TOF determination in talc and kaolin powder. TOF and TIF fractions (%) are calculated relative to TF measured using the *Final optimised combustion method*. TF and TIF were determined in triplicate ($n = 3$). TOF was determined by subtracting TIF from the mean TF.

a pre-combustion step at 525 °C for 2 h, the approach enabled selective quantification of organic fluorine while minimising interference from inorganic fluorides.

The method was assessed using a broad range of fluorinated compounds, including polymeric and non-polymeric species, as well as inorganic salts with diverse thermal properties. Optimised pre-combustion conditions ensure complete volatilisation of organic fluorine, even for highly stable compounds such as PTFE. Comparative analysis with aqueous extraction revealed significant discrepancies, particularly in complex matrices such as mineral fillers. Aqueous extraction underestimated TIF, leading to overestimation of TOF and potential false positives. In contrast, the pre-combustion approach provided more consistent and reproducible results, making it the most suitable among those evaluated for regulatory applications, while acknowledging that no method is entirely free from limitations. Indeed, certain limitations remain as highly refractory inorganic fluorides such as CaF_2 cannot be completely volatilisation, limiting the determination of absolute TF values. Conversely, certain volatile inorganic salts (e.g. NH_4BF_4) may partially or fully volatilise during pre-combustion, leading to underestimation of TIF and overestimation of TOF. These limitations highlight the need for cautious interpretation, particularly in heterogeneous matrices with unknown fluorine speciation. Nevertheless, the optimised pre-combustion method offers clear advantages: it minimises false positives, improves accuracy in distinguishing organic from inorganic fluorine, and supports compliance with regulatory thresholds. Its robustness, reproducibility, and adaptability make it a strong candidate for harmonised standards and routine monitoring of TOF as proxy to control PFAS presence in FCMs.

The study also demonstrated that mineral fillers commonly used in paper manufacturing (e.g. talc, kaolin, calcium carbonate) may contain fluorine impurities that significantly affect TOF measurements. Careful selection of pre-combustion conditions is therefore critical to avoid overestimation of TOF, which could otherwise lead to misinterpretation. Under the EU PPWR, the regulatory limit of 50 mg/kg applies to TF; if TF exceeds this threshold, it becomes essential to demonstrate that the detected fluorine is not attributable to perfluorinated organic compounds. Additionally, Denmark has established a guideline of 20 mg/kg for TOF in paper and board FCMs without functional barriers, further emphasising the need for reliable differentiation between organic and inorganic fluorine.

Future work should extend the application to a wider range of real FCMs, including multilayer packaging, recycled paper, and coated surfaces, to confirm robustness across industrial contexts. Integration into European standardisation efforts (CEN, ISO) could support harmonised TOF determination procedures and regulatory enforcement, provided these constraints are addressed. Further research should also investigate fluorine

content in industrial additives and mineral fillers, given their prevalence in paper and packaging materials. Beyond the scope of this study, recent work emphasises the importance of combining sum-parameter approaches (TF/TOF) with targeted and non-targeted mass spectrometry to achieve a comprehensive assessment of fluorine contamination. While LC/MS/MS was not performed here, previous studies consistently report that targeted PFAS account for only a minor fraction of total fluorine, often less than 1% (Schultes et al. 2019; Jovanović et al. 2024). This finding highlights a potential gap between TF values and the amount of fluorine quantified by targeted analysis, underscoring the need for complementary strategies: using TF/TOF to quantify global fluorine content and applying targeted/non-targeted MS to identify specific contributors where feasible. Such integration is essential for regulatory reliability in complex matrices containing both inorganic fluorine and diverse PFAS classes. Coupling this method with high-resolution mass spectrometry may enhance identification of unknown fluorinated compounds, bridging the gap between targeted and non-targeted analytical approaches. Ongoing research efforts are addressing this gap by implementing comprehensive workflows that include TF, TOF, EOF, and non-targeted/suspect LC-HRMS screening, providing a more holistic understanding of fluorine contamination in food contact materials.

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Disclosure statement

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